ABSTRACT

SYSTEM ARCHITECTURE DESIGN FOR AUTOMATED CONTAINER INSPECTION IN PORTS

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Container inspection processes at ports still face information fragmentation among stakeholders, inspection records that are not yet connected to container history, and limited visual evidence that can be referenced later. These conditions create a risk that container identity, inspection results, and inspection evidence are scattered across different media. This final project aims to design a container inspection automation system architecture that supports container identification, inspection result recording, visual evidence management, and container history presentation within a single information structure.

The design was conducted using the Design Science Research approach, which includes problem identification, objective definition, artifact design, demonstration through system realization, and result evaluation. The system architecture was structured using a layered approach consisting of presentation, application, data, and edge layers. In this design, the edge component handles video stream acquisition and initial inference, the API service manages orchestration and data persistence, and the web application provides an interface for monitoring inspection results. From the data perspective, the container entity is positioned as the central link between identity, inspection results, manifest information, and visual evidence.

The system realization shows that the proposed architecture can be implemented as a flow connecting the edge component, API service, data storage, and web interface. The evaluation was carried out through integration testing of the main flows, including the creation of a container entity through OCR, propagation of the container number to other scanning flows, storage of inspection results in the core data model, and presentation of results in the web application. Based on these evaluation results, the proposed design supports container history recording within the scope of the internal realization of this final project. The main contribution of this final project is an architecture design that maps responsibility separation among components, places the container entity as the center of inspection information, and evaluates the suitability of the design against the system realization.

Keywords: system architecture, container inspection automation, data integration, port, container history.